ANTTI KOLJONEN

INTELLIGENT SYSTEMS ENGINEERING STUDENT

Kajaani, Finland | contactme@ar-koljonen.fi | ar-koljonen.fi | github.com/Rumiraj

PROFESSIONAL SUMMARY

Intelligent Systems Engineering student at Kajaani University of Applied Sciences, focusing on artificial intelligence, embedded systems and software development. Builds practical applications using Python, C/C++, Java, computer vision and microcontrollers. Brings over 10 years of work experience across industrial manufacturing, metalworking, networking and customer-facing roles.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, HTML, CSS

AI and computer vision: PyTorch, OpenCV, ResNet18 transfer learning, TensorFlow coursework, NVIDIA CUDA

Embedded systems: ESP32, ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, sensors, embedded Linux / Yocto

Software and infrastructure: Linux, Git, GitHub, GitLab, Android Studio, VS Code, Node.js, REST APIs, MongoDB, MariaDB

Networking and industrial: LAN and VoIP installation, Wireshark, machining, welding, manufacturing processes

SELECTED TECHNICAL PROJECTS

VisualQC — AI-Powered Visual Inspection | *2026* | *Functional prototype*

- Built a camera-based manufacturing inspection prototype using Python, OpenCV and PyTorch, combining component detection with ResNet18 image classification.
- Prepared and labeled image data, trained a model using NVIDIA CUDA, and integrated inference into a live PASS/FAIL inspection workflow; further validation is ongoing.

IoT Environmental Monitoring System | *2026* | *In development*

- Developing a system connecting an ESP32 sensor device, Android application, Node.js/Express REST API and MongoDB database.
- Established Android-to-backend communication; ESP32 sensor transmission and automated data collection remain in development.

Engineering Portfolio Website | *2026* | *Published*

- Developed and deployed ar-koljonen.fi using HTML, CSS, GitHub Pages and custom-domain DNS configuration; maintains technical project documentation on GitHub.

Embedded Systems Coursework | *2025–2026*

- Implemented ESP32 FreeRTOS exercises involving tasks, mutexes and semaphores; worked with Arduino IMU data collection, sensor processing and Raspberry Pi embedded Linux / Yocto.

PROFESSIONAL EXPERIENCE

Network Technician — Digicenter Oy | *Finland* | *Within 2012–2021*

- Installed local networking equipment and VoIP communications devices; assisted with basic network design and technical installations.

Plater-Welder and Machinist — AC-Cranes Trading Oy | *Finland* | *Within 2012–2021*

- Worked in fabrication and machining for crane and lifting-equipment manufacturing, developing practical knowledge of industrial processes and mechanical systems.

Machine Operator — Sako Oy | *Finland* | *Within 2012–2021*

- Operated machinery in cartridge manufacturing and gained experience in structured industrial production environments.

Driver and Other Employment — Lähitaksi Oy, Multicab Oy, Dolmus Oy and others | *Finland* | *2012–2021 and alongside studies*

- Worked in passenger transport and additional customer-facing roles, building dependable service, communication and independent problem-solving skills.

Military Service — Finnish Defence Forces | *2013*

- Completed military service and reached the rank of Corporal.

EDUCATION

Bachelor of Engineering in Information and Communication Technology | *Kajaani University of Applied Sciences* | *Sep 2021–present*

Specialization in Intelligent Systems, including AI, embedded devices and programming. Expected graduation: early 2028.

Vocational Qualification in Metalworking | *HYRIA Vocational Institute, Hyvinkää* | *Sep 2009–Jun 2012*

Plater-Welder / Machinist.

ADDITIONAL INFORMATION

Driving licences: Categories A and B

Interests: Computer hardware, reverse engineering, mechanical projects, vehicle maintenance and outdoor activities

Team achievement: Member of Numen, a competitive World of Warcraft team with multiple world-first raid completions

References

Markku Karppinen - Teacher - 0447101373 - markku.karppinen@kamk.fi

Eero Huusko - Teacher - 0447157040 - eero.huusko@kamk.fi